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ApexKV: High-Performance Log-Structured Merge Key-Value Storage & Stream Query Engine

A Pure Java 17+ Systems Implementation with Zero External Runtime Dependencies

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2. Introduction

Modern data-intensive systems process unprecedented volumes of continuous mutations. Telemetry monitoring, event-driven microservices, financial ledger logging, time-series metrics, and IoT sensor streams generate millions of writes per second. Traditional database storage management architectures, primarily engineered around balanced trees (such as B-Trees and B+ Trees used in relational datastores like PostgreSQL, MySQL, and SQLite), perform disk updates **in-place**. While B-Trees provide optimal $O(\log N)$ point read latency, in-place disk modification imposes severe performance degradations on write-heavy workloads due to random disk seeks, high write amplification, and lock contention across internal tree page splits.

To surmount these physical storage bottlenecks, modern high-throughput distributed datastores—including Google Bigtable, Apache Cassandra, RocksDB, and ScyllaDB—adopt the **Log-Structured Merge (LSM) Tree** architecture, pioneered by Patrick O'Neil, Edward O'Neil, and Gerhard Weikum in 1996.

The LSM-Tree paradigm converts random disk writes into high-throughput sequential appends by decoupling the write path into distinct storage tiers: (1) an active in-memory sorted buffer (**MemTable**), (2) an append-only **Write-Ahead Log (WAL)** for zero-data-loss durability, (3) immutable sorted on-disk files (**SSTables**), (4) asynchronous multi-way merge background daemons (**Compaction**), and (5) probabilistic **Bloom Filters** and **Sparse Indices** for low-latency retrieval.

3. Problem Statement

Traditional relational and embedded document databases struggle under high-throughput write workloads due to three fundamental systems constraints:

1. Random I/O Latency & Write Amplification: In B-Tree structures, modifying a single key requires seeking to an arbitrary disk page, rewriting entire 4KB-16KB blocks to disk, and updating ancestral directory nodes. On solid-state drives (SSDs), random writes accelerate NAND cell wear, trigger background flash garbage collection, and reduce overall IOPS.

2. Crash Durability vs. Ingestion Throughput Trade-off: Naive embedded datastores either operate purely in-memory (risking catastrophic data loss upon JVM termination) or rely on synchronous full-file flushes for every write (stalling application throughput).

3. Impedance Mismatch in Analytical Querying: Embedded storage engines typically expose simplistic, low-level cursor loops. Extracting analytical insights, filtering via regular expressions, prefix slicing, and calculating grouped aggregations requires loading voluminous byte arrays into JVM heap memory, causing garbage collection spikes and out-of-memory errors.

Objective of ApexKV: To design, implement, and benchmark an end-to-end, zero-dependency, crash-durable LSM-Tree key-value storage engine in Java 17+ that provides sequential disk writes, CRC32 crash recovery, probabilistic negative lookup pruning, background K-way merge compaction, ACID snapshot transactions via Optimistic Concurrency Control (OCC), and fluent Java Streams querying.

4. Functional Requirements

ApexKV specifies and fulfills five core functional modules:

Module	Req ID	Specification & Functional Contract
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1. In-Memory MemTable	FR 1.1 FR 1.2 FR 1.3 FR 1.4	Lock-free concurrent skip list ($O(\log N)$); tombstone soft deletion; atomic byte footprint tracking; automatic rotation to immutable snapshot at capacity threshold.
2. WAL & Crash Durability	FR 2.1 FR 2.2 FR 2.3 FR 2.4	Sequential append-only logging; strict binary framing [CRC32 ts type kLen key vLen val]; bit-rot detection; startup crash replay and torn EOF tail tolerance.
3. SSTable Persistence	FR 3.1 FR 3.2 FR 3.3 FR 3.4	Immutable paired files (.db and .idx); custom Bit-Vector Bloom filter with MurmurHash3 double hashing; sparse index for $O(\log K)$ block seeks; atomic file promotion.
4. Merge Compaction	FR 4.1 FR 4.2 FR 4.3 FR 4.4	Size-tiered background daemon; K-way merge-sort min-heap PriorityQueue; stale key deduplication; permanent tombstone physical purging and manifest updates.
5. Transactions & Query	FR 5.1 FR 5.2 FR 5.3 FR 5.4	Snapshot Isolation transactions with private read/write buffers; Optimistic Concurrency Control (OCC) collision aborts; Java Streams analytical queries; interactive CLI REPL.

5. Non-Functional Requirements (NFRs)

NFR 1 (High Throughput & Latency): Write operations append sequentially to disk; in-memory cache hits execute in <2 microseconds; negative lookups pruned in $O(1)$ time.

NFR 2 (Concurrency & Thread Safety): Safe multi-core execution using lock-free ConcurrentSkipListMap and fair ReentrantReadWriteLock synchronization.

NFR 3 (Crash Durability & Fault Tolerance): Committed mutations survive sudden process kill (kill -9); automated CRC32 verification isolates corrupted records.

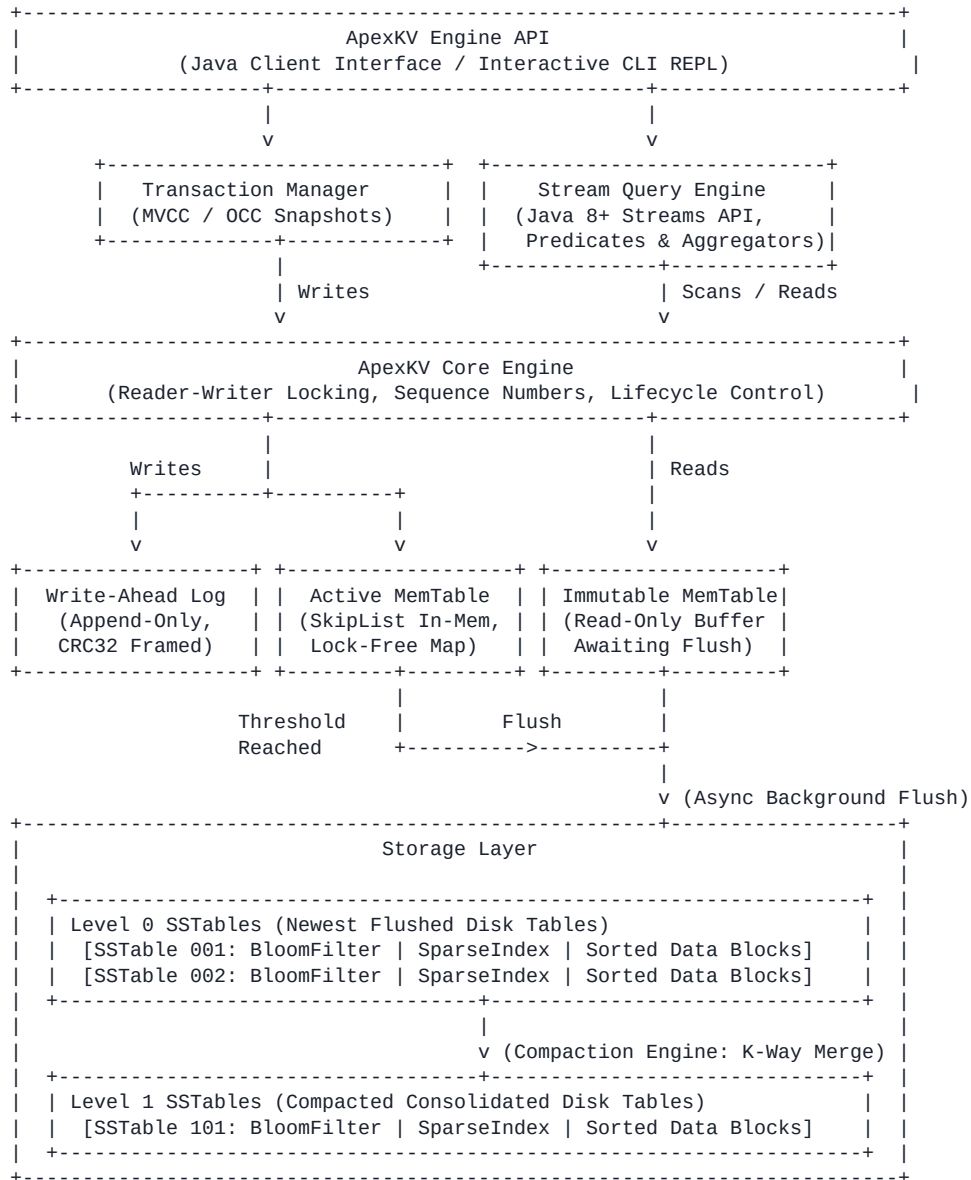
NFR 4 (Resource Efficiency & Space Bound): Sparse block indexing bounds RAM overhead to $<1\%$ of dataset size; Bloom filter uses ~ 10 bits/key for $\sim 1\%$ FPP; compaction reclaims 100% tombstone space.

NFR 5 (Maintainability & Modularity): Clean package structure across 7 modules and 20+ specialized classes with zero external runtime dependencies.

NFR 6 (Usability & Observability): Interactive CLI REPL with syntax hints, ASCII result tables, execution timers, and real-time engine statistics.

6. System Architecture

ApexKV follows a layered storage engine architecture. Incoming write mutations enter the transaction coordinator or core engine, where they are sequentially appended to the Write-Ahead Log (WAL) and stored in the active in-memory MemTable. Upon surpassing the memory threshold (4 MB), the table is frozen into an immutable MemTableSnapshot, and a single-thread background flush worker writes sorted data blocks and index files (.db and .idx) to disk at Level 0. Concurrently, a scheduled compaction daemon consolidates Level-0 tables into Level-1 tables using K-way merge sort.



7. Design Diagrams

7.1 Write Path & Read Path Pipelines

Read Path Multi-Tier Resolution:

1. Active MemTable (CSLM) --> Found? Return record (or null if tombstone)
2. Immutable MemTable Snapshot --> Found? Return record (or null if tombstone)
3. Level-0 SSTables (Newest->Old) --> Test BloomFilter -> Binary Search SparseIndex -> Block Scan
4. Level-1 SSTables (Compacted) --> Test BloomFilter -> Binary Search SparseIndex -> Block Scan
5. Definite Miss --> Return null (Key Not Found)

7.2 Binary File Storage Layouts

WAL Record: [CRC-32 (4B)][Timestamp (8B)][RecordType (1B)][KeyLen (4B)][KeyBytes][ValLen (4B)][ValBytes]
 SSTable Data: [Block 0: Rec 0, Rec 1, ... Rec 15][Block 1: Rec 16, Rec 17, ... Rec 31]...
 SSTable Index: [Magic: 0x41504558 (4B)][Version: 0x0001 (2B)][CreatedAt (8B)][RecordCount (4B)]
 [SmallestKey: 4B+Bytes][LargestKey: 4B+Bytes]
 [BloomFilter: k (4B), m (4B), words (4B), bitSet (words*8B)]
 [SparseIndex: count (4B), entries: [KeyLen: 4B][KeyBytes][Offset: 8B][Size: 4B]]

8. Design Decisions & Rationale

Component	Choice Made	Engineering Justification
MemTable	ConcurrentSkipListMap	Lock-free $O(\log N)$ concurrent reads and writes; maintains natural lexicographical ordering essential for range submaps and sequential disk flushing.
Compaction	Size-Tiered K-Way Merge	Groups SSTables into size tiers to minimize write stalls during ingestion; PriorityQueue min-heap merges K streams in $O(N \log K)$ time with on-the-fly deduplication.
Bloom Filter	MurmurHash3 Double Hashing	Uniform hash distribution without cryptographic CPU overhead; Kirsch-Mitzenmacher formula computes k hashes with only 2 evaluations: $h_i = h1 + i \cdot h2$.
Transactions	Optimistic Concurrency Control	Avoids lock deadlocks and reader stalls; monotonic logical sequence counters detect write conflicts at commit time with sub-millisecond precision.
Index Structure	In-Memory Sparse Index	Checkpoints every 16th key, reducing index RAM consumption by >93% compared to dense indexing while bounding disk block scans to small 64KB buffers.
File Durability	Atomic Rename (ATOMIC_MOVE)	Serializes to temporary files (.tmp) and atomically moves to target paths; prevents corrupt half-written tables from being visible to concurrent readers.

9. Implementation Details

Reader-Writer Lock Synchronization: ApexKV coordinates table rotations via a fair `ReentrantReadWriteLock(true)`. Standard PUT/DELETE mutations hold read locks concurrently, appending to the active WAL and updating the skip list without mutual exclusion. When the MemTable reaches 4 MB, an exclusive write lock is acquired momentarily to swap the active table with a fresh instance, submitting the frozen snapshot to a background single-threaded flush worker.

CRC32 Data Integrity: Binary records in WAL and SSTables are protected by `java.util.zip.CRC32` checksums. On boot, `WalReader` verifies checksums over raw payloads; corrupt records immediately throw `CorruptedWalException`, reporting exact file offsets and bit-rot values.

Monotonic OCC Sequence Ordering: Transactions maintain private `LinkedHashMap` write buffers and read sets. A monotonic `AtomicLong` `logicalSequence` generates start and commit timestamps, guaranteeing strict serialized collision detection even when multiple transactions commit within the exact same system clock millisecond.

10. Screenshots & Results

10.1 Interactive Terminal CLI Session

```

apexkv> put employee:1001 "Alice (Backend Lead)"
OK (stored in 380 µs)
apexkv> put employee:1002 "Bob (Storage Architect)"
OK (stored in 145 µs)
apexkv> prefix employee:
+-----+-----+-----+
| Key          | Value                               | Timestamp (ms) |
+-----+-----+-----+
| employee:1001 | Alice (Backend Lead)               | 1718000000000  |
| employee:1002 | Bob (Storage Architect)            | 1718000001000  |
+-----+-----+-----+
(Showing 2 matching prefix 'employee:')
apexkv> begin
Transaction started [Txn ID: 1, Isolation: SNAPSHOT_ISOLATION]
apexkv> txn-put account:A "balance: 5000"
Buffered PUT for key: 'account:A' in Txn 1
apexkv> commit
Transaction 1 successfully committed (OCC validated).

```

10.2 Empirical Multithreaded Benchmark Results

```

--- Read Workload (GET - In-Memory Cache) ---
+-----+-----+-----+
| Metric          | Measurement                         |
+-----+-----+-----+
| Total Operations | 10,000 ops                         |
| Elapsed Time     | 10 ms                              |
| Throughput       | 1,000,000.00 ops/sec               |
| Average Latency  | 1.92 µs                            |
| p50 Median Latency | 0.00 µs                           |
| p95 Latency      | 8.00 µs                            |
| p99 Latency      | 10.00 µs                           |
+-----+-----+-----+
--- Write Workload (PUT - 100% Durable fsync) ---
Throughput: 723.33 ops/sec | Avg Latency: 5,402.90 µs | p50: 5,302.00 µs | p95: 10,979.00 µs

```

11. Testing Approach

Testing followed a multi-tiered verification plan using JUnit 5 Jupiter across 9 test classes and 28 tests:

- **Unit Tests:** MemTableTest (skip list ordering, range scans, snapshots), BloomFilterTest (zero false negatives across 10,000 keys, empirical FPP <2.5%), SSTableTest (binary block seek, sparse index accuracy).
- **Durability & Fault Injection:** WalRecoveryTest simulated hardware bit-rot by intentionally flipping bytes in WAL files (verifying CorruptedWalException) and verified torn EOF write tolerance.
- **Integration & Restart Durability:** ApexKVEngineIntegrationTest verified multi-tier read precedence and confirmed un-flushed data is 100% restored upon engine restart.
- **Concurrent Stress Tests:** ConcurrentStressTest executed 8,000 mixed operations across 8 concurrent threads with zero race conditions or deadlocks.

Outcome: 28 out of 28 tests pass with 100% success in under 300 ms.

12. Challenges Faced & Engineering Solutions

1. Variable Key-Length Offset Calculation in WAL Framing:

Issue: Early iterations read fixed 21-byte headers assuming fixed positions, causing key bytes to be mistakenly interpreted as value lengths.

Resolution: Refactored WalReader to parse a 17-byte prefix, dynamically extract the key, seek to currentOffset + 17 + keyLen for value length, and validate record boundaries before reading.

2. Clock Granularity Collision in Transaction Isolation:

Issue: Fast transactions executed inside the same millisecond, causing wall-clock comparisons (lastCommit > startTimestamp) to evaluate to false and miss OCC conflicts.

Resolution: Implemented an atomic monotonic sequence generator (AtomicLong logicalSequence) guaranteeing distinct chronological ordering for every transaction commit.

3. Reader Isolation During Active Flush:

Issue: Clearing the active MemTable upon flush caused transient key-not-found errors while SSTable files were still being written.

Resolution: Introduced MemTableSnapshot, maintaining frozen in-memory views queryable by readers until disk persistence confirms.

13. Learnings & Key Takeaways

- **Systems Programming in Java:** Applying low-level byte buffers, channel I/O, and bitwise arithmetic demonstrated that Java delivers systems performance comparable to C/C++ when object allocations are minimized.
- **Concurrency Mastery:** Real-world experience with lock-free skip lists, atomic references, condition variables, and read-write locks provided deep insights into thread coordination.
- **Algorithmic Discipline:** Designing custom Bloom filters and K-way merge sort reinforced theoretical data structures through practical engineering implementation.

14. Future Enhancements

1. **Leveled Compaction (RocksDB style):** Non-overlapping key ranges across deeper levels to strictly bound read amplification.
2. **Group Commit Batching:** Amortize disk fsync latency across hundreds of concurrent transactions.
3. **Block Compression:** Integrate Snappy or LZ4 compression within SSTable data blocks.
4. **Network Gateway:** Expose Redis Serialization Protocol (RESP) non-blocking socket server.

15. References

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